NIAMH COLES

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EDUCATION

Texas A&M University *B.S. Multidisciplinary Engineering Technology* Aug 2023 - May 2027

Austin Community College - Dual Credit / TAMU Engineering Academy | Aug 2019 - May 2024

Relevant Coursework: Mechatronics I, Embedded Systems Development in C, Microcontroller Architecture, Analog Electronics, Electronic Instrumentation, Applied Dynamic Systems

TECHNICAL SKILLS

CAD & Simulation: SolidWorks, Fusion 360, OnShape; FEA (linear & non-linear static), dynamic modeling, MATLAB/Simulink, rapid prototyping

Electronics & PCB: Altium Designer & rigid/flex PCB design, custom sensor design & evaluation, soldering

Lab & Test: Oscilloscopes, multimeters, bench power supplies, CAN bus debugging, hardware bring-up, schematics & mechanical drawings

Programming: C, C++, Python, Assembly, Arduino, Git, LaTeX

Controls & Robotics: PID & feedforward control, robotics control systems, tendon-driven actuation

Professional: Design of experiments (DOE), technical presentations, technical documentation, literature review, mentoring & teaching

RELEVANT EXPERIENCE

Undergraduate Researcher - *Faculty Research Lab, Texas A&M* Jan 2025 - Present

- Independently designed and built a multimodal rigid/flex PCB in Altium Designer integrating temperature, humidity, acceleration, vertical force, and shear detection, driven by a Nordic nRF5340 (BLE) microcontroller module.
- Built a low-cost proof-of-concept shear-warning device using force-sensitive resistors (FSRs); initial results detect on-axis shear of ~5 N for as little as $20 while measuring vertical force to within 1-2 N.
- Designed validation experiments generating Bland-Altman plots for the temperature and humidity sensors, and built Python tools to analyze data and produce figures of merit supporting design conclusions; preparing a first-author submission to IEEE Sensors.

Research Experience for Undergraduates (REU) - *HAND ERC* Jun 2025 - Aug 2025

- Built the actuation and control system for a backdrivable, n+1 underactuated, four-DOF tendon-driven robotic finger with a four-bar coupling on two flexion joints.
- Implemented PID and gravity feedforward control across four DC motors with external magnetic encoders, using motor torques and Jacobians to drive the finger to commanded positions; built a Python UI for real-time monitoring and PID tuning.
- Designed and fabricated motor mounts and wired/soldered the full electronics system; presented progress bi-weekly.
- Diagnosed a CAN bus communication failure using an oscilloscope, CAN sniffer, and continuity testing; documented the process in a CAN debugging guide for future users of the system.
- Wrote a startup calibration routine that zeroes encoder offsets at a known reference pose; documented calibration drift from heat-softened motor mounts and other failure modes, with recommended fixes.

Student Worker - *MXET Program, Texas A&M* Jun 2026 - Jul 2026

- Designed and taught four weeks of curriculum for the HAND ERC Research Experience for Teachers, guiding six high-school teachers in building a simplified five-DOF tendon-driven robotic hand.
- Covered CAD, C++/Arduino, basic circuits, 3D-printing design constraints, and tendon-driven robot design; every teacher independently designed, assembled, and programmed a working hand, most with little prior experience.

PROJECTS & LEADERSHIP

Dexterous Design Challenge (DDC) - Co-Founder & Organizer - *HAND ERC* Jan 2026 - May 2026

- Co-founded and led an 8-week competition where undergraduate teams built 11-DOF robotic hands and wrists (113 participants, 20 teams, 19 majors); designed the forearm and build kit and co-wrote the rulebook and benchmarks.
- Organized informationals, kit distribution, check-ins, and communications, and helped plan and run the two-day final competition (graduate-student judges, high-school volunteers, evaluation testbeds, catering).
- First author on a resulting paper presented at an international IEEE conference (ICELIE 2026, Oct 18-21); contributed to technical work and manuscript writing.

Vice President & Technical Lead - *ASME-Tech, Texas A&M* Aug 2025 - Present

- Lead the technical team and two junior officers; plan and run build events, workshops, and study sessions for groups of 10-60 peers.

20-DOF Mechanical Hand Prototype - *Personal Project* May 2026 - Present

- Designed a 20-DOF hand with five splay DOF and three flexion/extension DOF per finger and thumb.
- Optimized low-friction joints and n+1 tendon routing channels for future tendon-based actuation; rapid-prototyped versions to verify joint fits and tendon channel tolerances.

3-DOF Spring-Loaded Finger - *Team Project, Applied Dynamic Systems* 2025

- Derived equations of motion and validated a Simulink/Simscape model against camera-tracked joint angles extracted in Python; traced error to servo current noise and friction.